Stack order

A stack follows Last In, First Out (LIFO). Push adds an item at the top, pop removes the top item, and peek reads the top item without removing it.

Push and overflow

For this array stack, capacity is 5 and top begins at -1. A push when top is 4 reports overflow and changes nothing. Otherwise increment top and store the new item at that index.

Pop and underflow

The array stack is empty when top is -1. Pop on an empty stack reports underflow without reading the array. Otherwise pop returns the item at top and then decrements top.

Checking brackets

To check bracket nesting, scan left to right. Push opening brackets. For a closing bracket, require a matching opening bracket at the stack top and pop it. Reject a mismatch or an empty stack at a closing bracket. At the end, accept only if the stack is empty.

Postfix operands

In postfix evaluation, push each numeric operand. At a binary operator, pop the right operand first and the left operand second. Compute left operator right, then push the result. This order matters for subtraction and division.